

DAVAO FRANCISCO BANGOY INTL, Philippines

WMO: 987530

Lat: 7.126N Lon: 125.646E Elev: 96 StdP: 14.64 Time Zone: 8.00 (E08) Period: 94-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.1	73.7	68.0	103.2	80.8	69.5	108.8	80.1	18.9	78.9	15.1	85.0	4.2	0	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	14.5	93.3	80.2	92.2	80.1	91.5	80.1	82.8	90.1	82.1	89.5	81.7	88.9	6.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	160.4	88.3	80.2	157.4	87.8	79.2	152.3	86.5	46.9	90.2	46.2	89.6	45.5	89.7	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.0	13.0	11.2	DB	70.0	96.3	2.7	2.1	68.1	97.8	66.5	99.0	65.0	100.2	63.1	101.7
			WB	67.5	84.7	2.9	1.1	65.4	85.5	63.7	86.1	62.1	86.7	60.0	87.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.7	81.2	81.7	82.7	83.9	83.9	83.0	82.5	82.8	82.9	83.3	82.8	82.4
	DBStd	1.84	2.22	1.91	1.92	1.68	1.59	1.53	1.46	1.46	1.37	1.26	1.61	1.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11953	966	886	1013	1018	1050	989	1009	1017	986	1032	984	1003
	CDD65	6478	501	466	548	568	585	539	544	552	536	567	534	538
	CDH74	69299	4662	4526	5753	6545	6715	5925	5863	6069	5866	6198	5592	5585
CDH80	26348	1521	1578	2212	2765	2736	2260	2138	2292	2274	2496	2074	2002	
Wind	WSAvg	5.1	5.9	6.4	6.1	5.5	4.8	4.3	4.4	4.8	4.4	4.5	4.6	5.3
Precipitation	PrecAvg	71.1	4.6	5.0	3.6	6.4	8.5	6.4	6.6	6.9	7.7	6.0	5.4	4.0
	PrecMax	88.7	9.8	10.2	7.9	12.8	18.2	12.4	15.3	13.7	13.0	9.0	9.7	8.9
	PrecMin	44.4	0.2	0.8	0.6	1.1	1.9	0.8	1.3	1.5	2.4	2.9	1.6	1.1
	PrecStd	11.5	2.6	3.1	2.3	3.7	4.2	3.4	3.7	3.3	3.3	1.7	2.4	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.7	92.0	93.6	94.9	93.9	92.9	91.7	91.9	92.9	93.3	92.9	92.4
		MCWB	79.4	78.9	78.6	79.3	80.7	81.6	80.4	80.2	80.8	81.2	80.6	80.3
	2%	DB	89.9	91.0	91.8	93.2	92.1	91.3	90.0	90.5	91.3	91.7	91.5	91.2
		MCWB	79.0	78.7	78.6	80.0	80.8	81.1	80.1	80.0	80.6	80.6	80.3	80.0
	5%	DB	89.2	89.6	91.0	91.7	91.2	89.8	89.4	89.6	89.8	90.6	90.0	89.7
		MCWB	78.8	78.5	78.6	80.0	80.7	80.3	80.0	79.9	80.0	80.3	80.0	79.6
	10%	DB	87.5	87.9	89.4	90.5	89.6	88.5	88.0	88.1	88.6	89.5	89.2	88.2
		MCWB	78.7	78.5	78.7	79.8	80.4	80.1	79.8	79.7	79.8	80.0	79.8	79.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	81.8	82.2	83.1	83.5	83.2	82.4	82.5	82.6	83.1	82.7	82.5
		MCDB	88.9	89.2	90.5	91.3	90.9	90.2	89.3	89.2	89.8	91.0	90.2	90.0
	2%	WB	80.6	80.6	81.0	81.9	82.4	81.9	81.4	81.5	81.6	81.8	81.7	81.4
		MCDB	87.6	87.9	88.8	90.0	89.6	88.8	88.1	88.1	88.9	89.4	89.1	88.6
	5%	WB	79.7	79.7	80.1	81.1	81.6	81.2	80.7	80.7	80.9	81.1	80.9	80.6
		MCDB	86.4	86.8	87.6	88.9	88.5	87.8	87.2	87.2	87.9	88.5	88.0	87.5
	10%	WB	79.0	78.9	79.3	80.4	80.9	80.5	80.0	80.0	80.2	80.4	80.1	79.8
		MCDB	85.4	85.8	86.5	87.9	87.5	86.9	86.2	86.3	86.8	87.6	86.8	86.4
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.3	14.2	14.5	13.4	12.9	12.6	12.9	13.5	14.3	13.5	13.1
		MCDBR	15.0	15.7	16.4	16.0	14.9	14.2	13.9	14.0	14.6	15.6	15.1	15.0
	5% WB	MCWBR	6.5	6.8	6.9	6.4	6.3	6.4	6.4	6.4	6.8	7.0	6.7	6.5
		MCDBR	13.3	14.0	14.5	14.7	13.7	13.3	13.0	13.2	13.8	14.6	14.2	14.0
		MCWBR	6.4	6.7	6.9	6.5	6.4	6.5	6.4	6.5	6.8	7.1	6.8	6.5
Clear-Sky Solar Irradiance	taub	0.407	0.404	0.403	0.412	0.410	0.399	0.406	0.424	0.426	0.446	0.409	0.406	
	taud	2.498	2.489	2.500	2.468	2.491	2.531	2.501	2.431	2.413	2.358	2.490	2.501	
	Ebn at Noon	288	292	292	284	279	279	278	277	280	276	285	286	
	Edh at Noon	35	36	36	36	35	33	34	37	38	40	35	34	
All-Sky Solar Radiation	RadAvg	1342	1547	1696	1772	1669	1564	1557	1618	1666	1608	1469	1387	
	RadStd	137	148	143	121	85	70	87	85	90	100	123	106	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	+0.43	+0.49	N/A	+0.74	+0.81	+0.72	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page